A Two-Tier Quotient Construction for $\mathcal{ALCI}_{RCC5}$

Period descriptors and PP-kernel nodes
for the finite representation of infinite models

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Abstract

We investigate model-theoretic quotient constructions for $\mathcal{ALCI}_{RCC5}$, a description logic extending $\mathcal{ALCI}$ with an RCC5 role box. The logic lacks the finite model property: the concept $C_\infty = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ is satisfiable only in infinite models containing unbounded PP-chains. We propose a *two-tier quotient* that represents such infinite models finitely. **Tier 1** (within-chain): each infinite PP-chain's eventually periodic tail is captured by a finite *period descriptor*—a cyclic word of Hintikka types. **Tier 2** (between-chain): chains interact through *PP-kernel nodes* with reflexive PP-loops, whose algebraic properties we establish: reflexive PP is universally self-absorbing and fully composition-consistent with all RCC5 relations. Single-type PP-chains collapse to a single kernel node. Multi-type periodic chains require period descriptors because PP-transitivity ($\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$) forces a strict linear order on distinct nodes, ruling out the PP-cycles that periodic demand satisfaction would require. We prove the algebraic foundations rigorously and identify a precise conjecture—the *periodic decomposition property*—whose resolution would yield decidability of $\mathcal{ALCI}_{RCC5}$.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with a role box derived from the five jointly exhaustive and pairwise disjoint (JEPD) base relations of the Region Connection Calculus RCC5 [6]: DR, PO, EQ, PP, PPI. These relations are used as role names in existential and universal restrictions, with the RCC5 composition table constraining the interpretation: for any three domain elements d_1, d_2, d_3, the relation $\rho(d_1, d_3)$ must lie in $\mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$.

Wessel [1] introduced this logic and left its decidability as an open problem, observing that $\mathcal{ALCI}_{RCC5}$ lacks both the finite model property and the tree model property. Several recent approaches have been attempted:

- A *quasimodel method* [2] proves soundness (satisfiable concept $\Rightarrow$ quasimodel) but has a gap in completeness (quasimodel $\Rightarrow$ model).
- A *contextual tableau* [3] proves soundness (closed family $\Rightarrow$ model) but its completeness conjecture $\mathsf{FW}(C, N)$ has been shown false [4]: the recentering mechanism requires growing PP-chains inside fixed-width local states, which is impossible.

In this paper we take a model-theoretic approach. Rather than designing a decision procedure directly, we ask: *can every infinite $\mathcal{ALCI}_{RCC5}$ model be represented by a finite quotient structure from which the original model can be recovered?* If the answer is yes and the quotient has bounded size (computable from the input concept), decidability follows by enumeration.

The source of infinity in $\mathcal{ALCI}_{RCC5}$ models is well-understood: PP-chains. The concept $C_\infty = (\exists\mathsf{PP}.\top) \sqcap (\forall\mathsf{PP}.\exists\mathsf{PP}.\top)$ forces an infinite chain $d_0 \ \mathsf{PP} \ d_1 \ \mathsf{PP} \ d_2 \ \mathsf{PP} \ \cdots$ where each element must have a strict PP-successor. Our key observation is that such chains have *regular structure*: the sequence of Hintikka types along the chain is eventually periodic (by finiteness of the type space and monotone accumulation of $\forall\mathsf{PP}$-consequences).

We exploit this regularity through a *two-tier quotient*:

1. **Tier 1** (within-chain): The periodic tail of each PP-chain is represented by a *period descriptor*—a finite cyclic word $(\tau_1, \ldots, \tau_p)$ of Hintikka types. This is a purely syntactic object (a word, not an RCC5 graph) that captures the within-chain demand satisfaction.
2. **Tier 2** (between-chain): Each chain is represented by a single *PP-kernel node* with a reflexive PP-loop. Cross-chain interactions are encoded as atomic RCC5 edges between kernel nodes and regular (non-chain) nodes.

We prove the algebraic foundations rigorously:

- Reflexive PP is *universally self-absorbing*: $R \in \mathsf{comp}(\mathsf{PP}, R)$ and $R \in \mathsf{comp}(R, \mathsf{PP})$ for every base relation R, and $\mathsf{PP} \in \mathsf{comp}(R, \mathsf{inv}(R))$ for all R (Theorem 3.1).
- Single-type PP-chains collapse to one kernel node with full demand satisfaction (Theorem 5.6).
- Multi-type periodic chains *cannot* be collapsed to kernel nodes: PP-transitivity forces a strict linear order on distinct nodes, making PP-cycles impossible (Theorem 3.4).
- Period descriptors resolve this obstruction (Definition 5.1).

The paper identifies a precise conjecture—the *periodic decomposition property* (Conjecture 7.1)—whose resolution would complete the decidability argument.

Structure. Section 2 recalls $\mathcal{ALCI}_{RCC5}$ and the RCC5 composition table. Section 3 establishes the algebraic properties of PP. Section 4 analyzes PP-chain structure. Section 5 introduces period descriptors. Section 6 defines the two-tier quotient. Section 7 states the remaining conjecture. Section 9 concludes.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts of $\mathcal{ALCI}_{RCC5}$ are built from concept names A, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$. We work in negation normal form (NNF); inverse roles are absorbed: $\exists\mathsf{PP}^-.C \equiv \exists\mathsf{PPI}.C$. We adopt **strong EQ semantics** (EQ is interpreted as identity); weak EQ reduces to strong via TBox internalization [1].

An *interpretation* is $\mathcal{I} = (\Delta^\mathcal{I}, \cdot^\mathcal{I})$ where $\Delta^\mathcal{I}$ is a nonempty set and each concept name A is interpreted as $A^\mathcal{I} \subseteq \Delta^\mathcal{I}$. A total function $\rho^\mathcal{I} : \Delta^\mathcal{I} \times \Delta^\mathcal{I} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ assigns an RCC5 base relation to every pair, subject to:

(i) $\rho^{\mathcal{I}}(d,d) = \mathsf{EQ}$ for all d;

(ii) $\rho^{\mathcal{I}}(d,e) = \mathsf{inv}(\rho^{\mathcal{I}}(e,d))$ for all d,e;

(iii) $\rho^{\mathcal{I}}(d_1,d_3) \in \mathsf{comp}(\rho^{\mathcal{I}}(d_1,d_2), \rho^{\mathcal{I}}(d_2,d_3))$ for all d_1,d_2,d_3 (composition consistency).

Existential restrictions: $d \in (\exists R.C)^{\mathcal{I}}$ iff there exists $e \in \Delta^{\mathcal{I}}$ with $\rho^{\mathcal{I}}(d,e) = R$ and $e \in C^{\mathcal{I}}$. Universal restrictions: $d \in (\forall R.C)^{\mathcal{I}}$ iff for all e with $\rho^{\mathcal{I}}(d,e) = R$, $e \in C^{\mathcal{I}}$.

2.2 Closure, Hintikka types, and composition table

Given a concept C_0 in NNF, the *closure* $\mathsf{cl}(C_0)$ is the set of subconcepts of C_0 and their NNF-negations. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset satisfying the usual local conditions (for each $C \sqcap D \in \mathsf{cl}(C_0)$: $C \sqcap D \in \tau$ iff $C \in \tau$ and $D \in \tau$; etc.). We write $\mathsf{Tp}(C_0)$ for the set of Hintikka types over $\mathsf{cl}(C_0)$. Note $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

The *RCC5 composition table* $\mathsf{comp} : \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}^2 \to 2^{\{\mathsf{DR},\mathsf{PO},\mathsf{PP},\mathsf{PPI}\}}$ (omitting EQ, which is the identity for composition) is:

$\mathsf{comp}(R_1, R_2)$	DR	PO	PP	PPI
DR	{all}	{DR, PO, PP}	{DR, PO, PP}	{DR}
PO	{DR, PO, PPI}	{all}	{PO, PP}	{DR, PO, PPI}
PP	{DR}	{DR, PO, PP}	{PP}	{all}
PPI	{DR, PO, PPI}	{PO, PPI}	{PO, PP, PPI}	{PPI}

where "all" denotes $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ (with EQ excluded for distinct elements).

The *patchwork property* of RCC5 [5]: every path-consistent atomic RCC5 constraint network is globally consistent (realizable by regions in $\mathbb{R}^n$ for $n \geq 23$).

2.3 Absence of the finite model property

Proposition 2.1 (Wessel [1])**.** *The concept* $C_\infty = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ *is satisfiable in* $\mathcal{ALCI}_{RCC5}$ *but has no finite model.*

Proof sketch. In any model, the root element d_0 has a PP-successor d_1; by $\forall$PP-propagation, d_1 also satisfies $\exists \mathsf{PP}.\top$, so it has a PP-successor d_2; and so on. Since $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, all d_i are pairwise distinct (the chain is a strict partial order). □

3 Algebraic properties of PP

We establish the key algebraic facts about PP in the RCC5 composition table. Throughout, R ranges over $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ (the relations between distinct elements).

Theorem 3.1 (Universal self-absorption of PP)**.** *The relation* PP *is* universally self-absorbing*:*

(a) ***Left absorption:*** $R \in \mathsf{comp}(\mathsf{PP}, R)$ *for every* R*.*

(b) ***Right absorption:*** $R \in \mathsf{comp}(R, \mathsf{PP})$ *for every* R*.*

(c) ***Self-closure:*** $\mathsf{PP} \in \mathsf{comp}(R, \mathsf{inv}(R))$ *for every* R*.*

Proof. Direct verification against the composition table:

Part (a): $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\} \ni \mathsf{DR}$; $\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\} \ni \mathsf{PO}$; $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\} \ni \mathsf{PP}$; $\mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PPI}$.

Part (b): $\mathsf{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\} \ni \mathsf{DR}$; $\mathsf{comp}(\mathsf{PO}, \mathsf{PP}) = \{\mathsf{PO}, \mathsf{PP}\} \ni \mathsf{PO}$; $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\} \ni \mathsf{PP}$; $\mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PPI}$.

Part (c): $\mathsf{comp}(\mathsf{DR}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PP}$; $\mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PP}$; $\mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PP}$; $\mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} \ni \mathsf{PP}$. □

Corollary 3.2 (Composition-consistency of reflexive PP)**.** *Let k be a node with $\rho(k, k) = \mathsf{PP}$ (a reflexive PP-loop). For any node m with $\rho(k, m) = R$, all composition constraints involving the triple (k, k, m) are satisfied:*

(i) $R \in \mathsf{comp}(\mathsf{PP}, R)$ *(triple (k, k, m)): by Theorem 3.1(a).*

(ii) $\mathsf{inv}(R) \in \mathsf{comp}(\mathsf{inv}(R), \mathsf{PP})$ *(triple (m, k, k)): since* $\mathsf{inv}(R) \in \mathsf{comp}(\mathsf{inv}(R), \mathsf{PP})$ *by Theorem 3.1(b) applied to* $\mathsf{inv}(R)$*.*

(iii) $\mathsf{PP} \in \mathsf{comp}(R, \mathsf{inv}(R))$ *(triple (k, m, k)): by Theorem 3.1(c).* □

Remark 3.3 (Other reflexive relations)**.** Not every RCC5 base relation is self-absorbing. Testing the same three conditions for DR, PO, and PPI:

- DR: *fails* left absorption. $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, so $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$. A reflexive DR-loop is not composition-consistent with all external edges.
- PO: passes all three conditions. However, $\mathsf{PO} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\text{all}\}$—wait, that works. Actually, PO fails condition (c) for $R = \mathsf{PP}$: $\mathsf{PO} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$—no, it does contain PO. More carefully: the composition table check reveals that PO *does* pass all three conditions. But PO is not transitive ($\mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\text{all}\}$, not $\{\mathsf{PO}\}$), so it does not create strict-order obstructions. It is not relevant for chain collapsing because PO-chains are not forced by the logic.
- PPI: passes all three conditions (by symmetry with PP).

The unique role of PP (and PPI) is the combination of self-absorption *and* transitivity. This makes them simultaneously the cause of infinite chains and the enabler of kernel-node collapse.

Theorem 3.4 (PP-transitivity forces strict linear order)**.** *Let $G = (V, \rho)$ be a finite set of nodes with an atomic RCC5 labeling (JEPD, composition-consistent). If $\rho(a, b) = \mathsf{PP}$ and $\rho(b, c) = \mathsf{PP}$ for distinct a, b, c, then $\rho(a, c) = \mathsf{PP}$. Consequently:*

(a) The restriction of ρ to PP *is a strict partial order (irreflexive, asymmetric, transitive).*

(b) There is no PP*-cycle among distinct nodes: no sequence $v_1, \ldots, v_n$ of distinct nodes with $\rho(v_i, v_{i+1}) = \mathsf{PP}$ for $i = 1, \ldots, n-1$ and $\rho(v_n, v_1) = \mathsf{PP}$.*

Proof. $\rho(a, c) \in \mathsf{comp}(\rho(a, b), \rho(b, c)) = \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $\rho(a, c) = \mathsf{PP}$.

(a) Irreflexivity: $\rho(a, a) = \mathsf{EQ} \neq \mathsf{PP}$. Asymmetry: $\rho(a, b) = \mathsf{PP}$ implies $\rho(b, a) = \mathsf{PPI} \neq \mathsf{PP}$. Transitivity: shown above.

(b) If $v_1, \ldots, v_n, v_1$ were a PP-cycle, then by transitivity $\rho(v_1, v_n) = \mathsf{PP}$, hence $\rho(v_n, v_1) = \mathsf{PPI}$. But the cycle requires $\rho(v_n, v_1) = \mathsf{PP}$, contradicting JEPD (PP and PPI cannot both hold). □

4 Structure of PP-chains in models

4.1 Monotone accumulation along PP-chains

Definition 4.1 (PP-chain)**.** In a model $\mathcal{I}$, a PP-*chain* is a sequence $(d_i)_{i\geq 0}$ of domain elements with $\rho^{\mathcal{I}}(d_i, d_{i+1}) = \mathsf{PP}$ for all $i \geq 0$. By Theorem 3.4, all d_i are pairwise distinct and $\rho^{\mathcal{I}}(d_i, d_j) = \mathsf{PP}$ for $i < j$.

Lemma 4.2 (Monotone ∀PP-accumulation)**.** *Let $(d_i)_{i\geq 0}$ be a* PP*-chain and define*

$$F_i = \{C \in \mathsf{cl}(C_0) : \forall\mathsf{PP}.C \in \mathsf{tp}(d_i)\}.$$

Then $F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots$. Moreover, if $\forall\mathsf{PP}.C \in \mathsf{tp}(d_i)$ then $C \in \mathsf{tp}(d_j)$ for all $j > i$.

Proof. If $\forall\mathsf{PP}.C \in \mathsf{tp}(d_i)$ and $\rho^{\mathcal{I}}(d_i, d_{i+1}) = \mathsf{PP}$, then $C \in \mathsf{tp}(d_{i+1})$ by the semantics of ∀PP. Furthermore, if $\forall\mathsf{PP}.C \in \mathsf{cl}(C_0)$ is a formula, then typically $C \in \mathsf{cl}(C_0)$ as well. The crucial point is that $\forall\mathsf{PP}.(\forall\mathsf{PP}.C)$ need not be in $\mathsf{tp}(d_i)$—the ∀PP prefix does not replicate itself. However, if $\forall\mathsf{PP}.C \in \mathsf{cl}(C_0)$ has the property that $\forall\mathsf{PP}.C$ is among the ∀PP-consequences of d_i (i.e., $\forall\mathsf{PP}.(\forall\mathsf{PP}.C) \in \mathsf{tp}(d_i)$), then it propagates.

More precisely: define $F_i = \{C : \forall\mathsf{PP}.C \in \mathsf{tp}(d_i)\}$. If $\forall\mathsf{PP}.C \in \mathsf{tp}(d_i)$, then since $\rho(d_i, d_{i+1}) = \mathsf{PP}$, we have $C \in \mathsf{tp}(d_{i+1})$. Now, C may or may not equal some $\forall\mathsf{PP}.D$. If it does, then $\forall\mathsf{PP}.D \in \mathsf{tp}(d_{i+1})$, so $D \in F_{i+1}$. The set F_i grows monotonically because every element of F_i generates at least its own consequences in F_{i+1}.

Since $\mathsf{cl}(C_0)$ is finite, the sequence (F_i) stabilizes: there exists $N \leq |\mathsf{cl}(C_0)|$ such that $F_N = F_{N+1} = \cdots$. □

Definition 4.3 (Stabilized tail)**.** The *stabilized tail* of a PP-chain $(d_i)_{i\geq 0}$ is the subsequence $(d_i)_{i\geq N}$ where N is the stabilization index of Lemma 4.2.

4.2 Eventually periodic type sequences

Proposition 4.4 (Eventual periodicity)**.** *The sequence of Hintikka types $(\mathsf{tp}(d_i))_{i\geq 0}$ along a* PP*-chain is* eventually periodic*: there exist $N \geq 0$ and $p \geq 1$ such that $\mathsf{tp}(d_{N+i}) = \mathsf{tp}(d_{N+i+p})$ for all $i \geq 0$.*

Proof sketch. After ∀PP-stabilization (index N), the types $\mathsf{tp}(d_N), \mathsf{tp}(d_{N+1}), \ldots$ all share the same ∀PP-core. The remaining freedom comes from ∃-formulas and concept names in $\mathsf{cl}(C_0)$. Each d_i's type is determined by its ∃-demands and their witnesses.

On the stabilized tail, each element d_i's ∃PP-demands are satisfied by d_{i+1} (or elements further along the chain). The choice of type for d_{i+1} is constrained by:

1. The ∀PP-consequences (fixed after stabilization).
2. The ∃PP-demands of d_i that d_{i+1} must satisfy.
3. The ∃-demands of d_{i+1} itself (which must be satisfiable, possibly by creating further chain elements or off-chain witnesses).

Since there are finitely many types ($|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$) and the constraints are local (each step depends only on the current type and the stabilized core), the sequence of types must eventually repeat. Once a type repeats, the same local constraints produce the same successor type, yielding periodicity. □

Remark 4.5. The argument in Proposition 4.4 is not fully rigorous. The difficulty is that a type's $\exists$-demands may be satisfied by elements at varying distances along the chain, not just the immediate successor. A complete proof would need to account for the "lookahead" needed to find witnesses. We revisit this gap in Section 7.

4.3 External relation stabilization

Lemma 4.6 (External relation monotonicity)**.** *Let $(d_i)_{i\geq 0}$ be a* PP*-chain and $e \notin \{d_i\}$ an external element. Define $R_i = \rho^{\mathcal{I}}(e, d_i)$. Then:*

(a) $R_{i+1} \in \mathsf{comp}(\mathsf{inv}(\mathsf{PP}), R_i) = \mathsf{comp}(\mathsf{PPI}, R_i)$ *for all i.*

(b) Equivalently: $R_{i+1} \in \mathsf{comp}(R_i, \mathsf{PP})$ *(viewing the chain from e's perspective via the triple (e, d_i, d_{i+1})).*

(c) The allowed transitions are:

$$\begin{aligned}\mathsf{comp}(\mathsf{DR}, \mathsf{PP}) &= \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\},\\ \mathsf{comp}(\mathsf{PO}, \mathsf{PP}) &= \{\mathsf{PO}, \mathsf{PP}\},\\ \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) &= \{\mathsf{PP}\},\\ \mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) &= \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}.\end{aligned}$$

(d) The relation PP *is* absorbing*: once $R_i = \mathsf{PP}$, we have $R_j = \mathsf{PP}$ for all $j > i$.*

(e) The sequence (R_i) stabilizes within at most $|\mathsf{cl}(C_0)|$ steps.

Proof. (a) and (b): By composition consistency of the triple (e, d_i, d_{i+1}) with $\rho(d_i, d_{i+1}) = \mathsf{PP}$.

(c): Direct from the composition table.

(d): $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $R_i = \mathsf{PP}$ implies $R_{i+1} = \mathsf{PP}$.

(e): The possible values $\{R_i\}$ form a decreasing chain under the preorder induced by reachability through $\mathsf{comp}(\cdot, \mathsf{PP})$. The reachability structure is: $\mathsf{PPI} \to \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$; $\mathsf{DR} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$; $\mathsf{PO} \to \{\mathsf{PO}, \mathsf{PP}\}$; $\mathsf{PP} \to \{\mathsf{PP}\}$. Once PP is reached, the sequence is constant. Since there are only 4 possible values, stabilization occurs within at most 4 steps. □

5 Period descriptors

Definition 5.1 (Period descriptor)**.** A *period descriptor* for C_0 is a tuple $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ where $p \geq 1$ and each $\tau_i \in \mathsf{Tp}(C_0)$ is a Hintikka type, satisfying:

(V1) **Type consistency:** Each τ_i is a Hintikka type in $\mathsf{cl}(C_0)$.

(V2) **$\forall$PP-closure:** If $\forall\mathsf{PP}.C \in \tau_i$ for some i, then $C \in \tau_j$ for all $j \in \{1, \ldots, p\}$.

(V3) **$\forall$PPI-closure:** If $\forall\mathsf{PPI}.C \in \tau_i$ for some i, then $C \in \tau_j$ for all $j \in \{1, \ldots, p\}$.

(V4) **$\exists$PP-witness:** If $\exists\mathsf{PP}.C \in \tau_i$ for some i, then $C \in \tau_j$ for some $j \in \{1, \ldots, p\}$.

(V5) **Cross-chain coherence:** For any external type σ and any two positions i, j, if the stabilized relation from σ to τ_i is S_i and to τ_j is S_j, then $S_j \in \mathsf{comp}(S_i, \mathsf{PP})$.

The descriptor represents the infinite periodic PP-chain

$$\cdots\ \tau_1\ \mathsf{PP}\ \tau_2\ \mathsf{PP}\ \cdots\ \mathsf{PP}\ \tau_p\ \mathsf{PP}\ \tau_1\ \mathsf{PP}\ \tau_2\ \mathsf{PP}\ \cdots$$

Definition 5.2 (Stabilized core)**.** The *stabilized core* of a period descriptor $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ is

$$\mathsf{Core}(\mathsf{Desc}) = \bigcap_{i=1}^{p} \tau_i$$

—the set of concepts present in *every* type of the period.

Lemma 5.3 (Core properties)**.** *Let* $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ *be a valid period descriptor.*

(a) If $\forall\mathsf{PP}.C \in \tau_i$ *for any* i*, then* $C \in \mathsf{Core}(\mathsf{Desc})$ *(by (V2)).*

(b) If $\forall\mathsf{PPI}.C \in \tau_i$ *for any* i*, then* $C \in \mathsf{Core}(\mathsf{Desc})$ *(by (V3)).*

(c) $\mathsf{Core}(\mathsf{Desc})$ *is closed under all* $\forall\mathsf{PP}$ *and* $\forall\mathsf{PPI}$ *consequences that appear anywhere in the period.*

Proof. Direct from (V2) and (V3): these conditions require the consequent C to be present in *all* types of the period, hence in their intersection. □

Remark 5.4 (Why (V3) is needed)**.** The condition (V3) accounts for $\forall\mathsf{PPI}$-propagation backward along the chain. Since $\rho(d_{i+1}, d_i) = \mathsf{PPI}$, any $\forall\mathsf{PPI}.C \in \mathsf{tp}(d_{i+1})$ forces $C \in \mathsf{tp}(d_i)$. On a periodic chain, this means C must appear in all period types.

Remark 5.5 (Why (V4) suffices for within-chain demands)**.** The condition (V4) requires that every $\exists\mathsf{PP}.C$ demand in the period can be witnessed by some type in the period. On the infinite periodic chain $\ldots\tau_1\ \mathsf{PP}\ \tau_2\ \mathsf{PP}\ \cdots\ \mathsf{PP}\ \tau_p\ \mathsf{PP}\ \tau_1\ \mathsf{PP}\ \cdots$, each position i has infinitely many future positions $j > i$ with each period type. If $C \in \tau_j$, then the element at position j on the chain witnesses the demand $\exists\mathsf{PP}.C$ of the element at position i (since $\rho(d_i, d_j) = \mathsf{PP}$ by transitivity).

Note that non-PP demands ($\exists\mathsf{DR}.C$, $\exists\mathsf{PO}.C$, $\exists\mathsf{PPI}.C$) are *not* required to be satisfied within the chain. These demands are satisfied by off-chain elements (handled in Tier 2 of the quotient).

5.1 Single-type period descriptors

Theorem 5.6 (Single-type collapse)**.** *If* $\mathsf{Desc} = (\tau)$ *is a single-type period descriptor (i.e.,* $p = 1$*), then the entire infinite* PP*-chain collapses to a single* kernel node k *with:*

- *Type* $\mathsf{tp}(k) = \tau$.
- *Reflexive* PP*-loop:* $\rho(k, k) = \mathsf{PP}$.
- *Every* $\exists\mathsf{PP}.C$ *demand in* τ *is self-witnessed:* $\exists\mathsf{PP}.C \in \tau$ *implies* $C \in \tau$ *(by (V4) with* $p = 1$*), so the reflexive loop satisfies the demand.*
- *Composition-consistency of* $\rho(k, k) = \mathsf{PP}$ *with all external edges: by Corollary 3.2.*

Proof. (V4) with $p = 1$ requires: $\exists\mathsf{PP}.C \in \tau$ implies $C \in \tau$. So every PP-demand is self-satisfiable via the reflexive loop. Composition-consistency follows from Theorem 3.1. □

Example 5.7. $C_\infty = (\exists\mathsf{PP}.\top) \sqcap (\forall\mathsf{PP}.\exists\mathsf{PP}.\top)$ has a single-type period: $\tau = \{\exists\mathsf{PP}.\top, \forall\mathsf{PP}.\exists\mathsf{PP}.\top, \top\}$. Since $\top \in \tau$, the demand $\exists\mathsf{PP}.\top$ is self-witnessed. The entire infinite chain $d_0\ \mathsf{PP}\ d_1\ \mathsf{PP}\ \cdots$ collapses to one kernel node.

5.2 Multi-type period descriptors

Theorem 5.8 (PP-cycle impossibility for multi-type periods)**.** *Let* $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ *with* $p \geq 2$ *and* $\tau_i \neq \tau_j$ *for some* $i \neq j$*. If we attempt to represent this period by* p *distinct kernel nodes* $k_1, \ldots, k_p$ *with* $\rho(k_i, k_{i+1}) = \mathsf{PP}$ *(indices mod* p*), then:*

(a) By Theorem 3.4, the PP*-edges force a strict linear order:* $k_1 \ \mathsf{PP}\ k_2 \ \mathsf{PP} \ \cdots \ \mathsf{PP}\ k_p$*.*

(b) The closing edge $\rho(k_p, k_1) = \mathsf{PP}$ *(needed for the period to "wrap around") contradicts the linear order: by transitivity,* $\rho(k_1, k_p) = \mathsf{PP}$*, hence* $\rho(k_p, k_1) = \mathsf{PPI} \neq \mathsf{PP}$*.*

(c) Consequently, the last kernel k_p cannot *satisfy* $\exists\mathsf{PP}.C$ *demands that require a type present only in* τ_j *for* $j < p$*.*

Proof. (a) By induction on $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$.

(b) $\rho(k_1, k_p) = \mathsf{PP}$ (by transitivity of the chain) forces $\rho(k_p, k_1) = \mathsf{inv}(\mathsf{PP}) = \mathsf{PPI}$. But the cyclic period requires $\rho(k_p, k_1) = \mathsf{PP}$, which contradicts JEPD.

(c) The demand $\exists\mathsf{PP}.C \in \tau_p$ with $C \notin \tau_p$ but $C \in \tau_j$ for some $j < p$: in the infinite chain, this is satisfied by the next-period occurrence of τ_j, which is PP-above d_{N+p}. But in the kernel representation, $\rho(k_p, k_j) = \mathsf{PPI}$ (since $j < p$), not PP. The demand cannot be satisfied within the kernel graph. □

This is precisely why period descriptors are needed: they represent the periodic structure *as a word* (a cyclic sequence of types), not as an RCC5 graph. The word representation does not require PP-cycles and correctly handles demand satisfaction via condition (V4).

6 The two-tier quotient construction

Given a model $\mathcal{I} \models C_0$, we construct a finite quotient $\mathcal{Q}(\mathcal{I})$.

6.1 Identifying PP-chains

In the tree unraveling $\mathcal{I}^T$ of $\mathcal{I}$ (unraveling the $\exists$-witness structure into a tree), each infinite branch is a sequence of $\exists R$-witness steps. An infinite PP-chain in $\mathcal{I}^T$ is a maximal path along which every step is an $\exists\mathsf{PP}$-witness step.

Remark 6.1. Other $\exists$-demands ($\exists\mathsf{DR}.C$, $\exists\mathsf{PO}.C$, $\exists\mathsf{PPI}.C$) create *branches* off the main PP-chain. Only $\exists\mathsf{PP}$ demands extend the chain itself. This is because $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ forces the chain to continue in the PP-direction, while other relations create independent subtrees whose roots are related to the chain by DR, PO, or PPI.

6.2 Tier 1: Period descriptors for PP-chains

For each maximal PP-chain $(d_i)_{i \geq 0}$ in $\mathcal{I}^T$:

1. Compute the stabilization index N (Lemma 4.2).
2. Extract the type sequence $(\mathsf{tp}(d_N), \mathsf{tp}(d_{N+1}), \ldots)$ on the stabilized tail.
3. By Proposition 4.4, this sequence is eventually periodic with period $(\tau_1, \ldots, \tau_p)$.

4. Record the period descriptor $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$.
5. The finite prefix $d_0, \ldots, d_{N-1}$ is retained as *regular nodes*.

6.3 Tier 2: Kernel nodes and cross-chain edges

For each period descriptor $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$:

1. Create a single **PP-kernel node** k_{Desc} with:
   - Type $\mathsf{tp}(k_{\mathsf{Desc}}) = \mathsf{Core}(\mathsf{Desc}) = \bigcap_{i=1}^{p} \tau_i$.
   - Reflexive edge $\rho(k_{\mathsf{Desc}}, k_{\mathsf{Desc}}) = \mathsf{PP}$.
2. For each other kernel node k' or regular node m in the quotient, define the edge $\rho(k_{\mathsf{Desc}}, m)$ as the **stabilized relation**: the eventual value of $\rho^{\mathcal{I}}(d_i, e)$ as $i \to \infty$ along the chain (which exists by Lemma 4.6).
3. The $\exists$-demands of k_{Desc} are handled in two ways:
   - $\exists\mathsf{PP}$ **demands** in $\mathsf{Core}(\mathsf{Desc})$: satisfied internally by the period descriptor (condition (V4)). If $p = 1$, the reflexive loop witnesses them directly. If $p > 1$, the witness exists within the periodic chain.
   - **Non-PP demands** ($\exists\mathsf{DR}.C$, $\exists\mathsf{PO}.C$, $\exists\mathsf{PPI}.C$) in $\mathsf{Core}(\mathsf{Desc})$: satisfied by other kernel nodes or regular nodes in the quotient via the cross-chain edges.

6.4 Regular nodes

1. The finite prefix of each PP-chain (before stabilization) is represented by individual nodes with their original types and pairwise RCC5 relations.
2. Non-chain elements (witnesses for non-PP demands) are represented individually or collapsed by type equivalence.
3. Edges between regular nodes, and between regular nodes and kernel nodes, are inherited from the model (or from the stabilized relation for kernel nodes).

6.5 Size bound

Proposition 6.2 (Quotient size)**.** *The quotient $\mathcal{Q}(\mathcal{I})$ has size bounded by a computable function of $|C_0|$:*

(a) ***Period descriptors:*** $p \leq |\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$. *The number of distinct descriptors is at most $|\mathsf{Tp}(C_0)|^{|\mathsf{Tp}(C_0)|} \leq 2^{2^{O(|C_0|)}}$ (doubly exponential, but finite and computable).*

(b) ***Kernel nodes:*** *one per distinct period descriptor.*

(c) ***Regular nodes:*** *at most $|\mathsf{cl}(C_0)| \cdot |\mathsf{Tp}(C_0)|$ per chain (the prefix length is bounded by the stabilization index $N \leq |\mathsf{cl}(C_0)|$).*

(d) ***Cross-chain edges:*** *at most $O(n^2)$ where n is the total number of nodes, each from $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.*

Remark 6.3 (Complexity)**.** The doubly-exponential bound on the number of period descriptors may be an overcount. If the period length p is bounded by a polynomial in $|\mathsf{cl}(C_0)|$ (which we conjecture but have not proven), the total quotient size would be at most singly exponential, consistent with an ExpTime upper bound.

7 The remaining gap: periodic decomposition

The two-tier quotient construction converts an infinite model to a finite representation. For this to yield a decision procedure, we need two properties:

1. **Completeness** (every satisfiable concept has a valid quotient): every infinite $\mathcal{ALCI}_{RCC5}$ model admits a two-tier decomposition whose period descriptors satisfy (V1)–(V5) and whose cross-chain edges are composition-consistent.
2. **Soundness** (every valid quotient corresponds to a satisfiable concept): every two-tier quotient satisfying all validity conditions can be "unfolded" back into a genuine $\mathcal{ALCI}_{RCC5}$ model.

We have established (Section 3) that the algebraic prerequisites hold: reflexive PP is composition-consistent, and the stabilization argument (Section 4) provides the structural framework for completeness. However, several precise claims remain unproven.

Conjecture 7.1 (Periodic decomposition property)**.** For every satisfiable $\mathcal{ALCI}_{RCC5}$ concept C_0, every model $\mathcal{I} \models C_0$ admits a two-tier decomposition $\mathcal{Q}(\mathcal{I})$ such that:

(P1) Every infinite PP-chain in $\mathcal{I}$ has an eventually periodic type sequence.

(P2) The cross-chain kernel edges (stabilized external relations) are globally composition-consistent: for any three nodes a, b, c in the quotient (kernel or regular), $\rho(a, c) \in \mathsf{comp}(\rho(a, b), \rho(b, c))$.

(P3) The quotient has size bounded by $f(|C_0|)$ for some computable f.

7.1 Status of each component

(P1) Eventual periodicity. The argument of Proposition 4.4 is plausible but incomplete. The difficulty is that an element's type on the chain may depend on the types of its off-chain witnesses, which are themselves determined by the model structure. In the tree unraveling, the off-chain structure at position i is independent of position $i+1$, so the type at position i can vary based on which off-chain demands happen to be satisfiable. A complete proof would need to show that the satisfiable type patterns along the chain are constrained enough to force periodicity.

A promising approach: after $\forall\mathsf{PP}$-stabilization, the set of *feasible types* at each position is the same (determined by the stabilized core plus the available concept names). The local constraints ($\exists\mathsf{PP}$ demands at position i must be witnessed at some position $j > i$) create a finite-state dependency that can be modeled as a Büchi automaton. The accepting runs of this automaton are exactly the valid type sequences, and every infinite accepting run of a Büchi automaton is eventually periodic.

(P2) Cross-chain composition consistency. The stabilized external relations come from the model, where they are globally consistent. The question is whether the act of *quotienting* (replacing an infinite chain by a single kernel node) preserves composition consistency.

For triples involving *one* kernel node: guaranteed by Theorem 3.1 (universal self-absorption).

For triples involving *two* kernel nodes k_1, k_2 and a regular node m: the relations $\rho(k_1, m)$ and $\rho(k_2, m)$ are stabilized model relations, and $\rho(k_1, k_2)$ is also a stabilized model relation. Since these three stabilized relations were originally present in the model (at sufficiently deep chain positions), and the model is globally consistent, the triple is composition-consistent.

For triples involving *three* kernel nodes: the same argument applies—each pairwise relation is the stabilized model relation between sufficiently deep elements of the respective chains.

This argument is essentially correct but requires a careful formalization of "sufficiently deep" (there must exist a single depth threshold beyond which all three pairwise relations have simultaneously stabilized).

(P3) Bounded quotient size. The number of kernel nodes is bounded by the number of distinct period descriptors. The number of regular nodes is bounded by the stabilization prefix length times the number of chains. Both are computable from $|C_0|$. The main question is whether the period length p can be bounded by a computable function of $|C_0|$. Since $p \leq |\mathsf{Tp}(C_0)|$ (a repeating type forces the period to close), this holds with a singly exponential bound.

7.2 Comparison with other approaches

Approach	Sound	Complete	Gap
Quasimodel [2]	yes	gap	Henkin extension may fail
Contextual tableau [3]	yes	**no**	$\mathrm{FW}(C, N)$ is false [4]
Two-tier quotient (this paper)	gap	gap	Periodicity (P1) and soundness (unfolding)

The two-tier quotient has gaps in both directions, but they are arguably more tractable than the quasimodel extension gap. The completeness gap (P1) is a claim about model structure that could be verified by analyzing PP-chain dynamics. The soundness gap (unfolding a valid quotient back into a model) is closely related to the patchwork property of RCC5, which has been fully characterized [5].

8 Soundness direction: unfolding the quotient

We sketch how a valid two-tier quotient could be unfolded into a genuine $\mathcal{ALCI}_{RCC5}$ model, highlighting the key lemma needed.

8.1 Unfolding the period descriptor

Given a period descriptor $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$, construct the infinite PP-chain:

$$d_1 \ \mathsf{PP} \ d_2 \ \mathsf{PP} \ \cdots \ \mathsf{PP} \ d_p \ \mathsf{PP} \ d_{p+1} \ \mathsf{PP} \ \cdots$$

where $\mathsf{tp}(d_i) = \tau_{((i-1) \bmod p)+1}$. The chain elements are new, fresh domain elements. All $\exists\mathsf{PP}$ demands within the chain are satisfied by construction (condition (V4)).

8.2 Assigning cross-chain relations

The quotient specifies a single stabilized relation R between kernel k_{Desc} and each other node m. In the unfolded model, we must assign a *specific* relation $\rho(d_i, m)$ for each chain element d_i.

- For the stabilized tail: $\rho(d_i, m) = R$ for all i in the periodic part. This is consistent because $R \in \mathsf{comp}(R, \mathsf{PP})$ (Theorem 3.1(b)), so the relation does not change along the chain.

- For the finite prefix: the quotient's regular nodes already carry specific relations.

8.3 Global consistency: the key lemma

Lemma 8.1 (Cross-chain consistency, sketch)**.** *Let d_i and d_j be elements on two different* PP*-chains, both in their stabilized tails, with stabilized relations $\rho(d_i, m) = R$ and $\rho(d_j, m) = S$ to a common node m, and $\rho(d_i, d_j) = T$ (the stabilized inter-chain relation). Then:*

$$T \in \mathsf{comp}(R, \mathsf{inv}(S)) \quad \textit{and} \quad S \in \mathsf{comp}(\mathsf{inv}(R), T) \quad \textit{and} \quad R \in \mathsf{comp}(T, S).$$

Proof sketch. In the original model, there exist indices i_0, j_0 large enough that the three pairwise relations have all stabilized. At those indices, the triple (d_{i_0}, d_{j_0}, m) is composition-consistent (since the model is globally consistent). The stabilized relations in the quotient inherit this consistency. □

The difficulty is formalizing "all three stabilize simultaneously" and showing that the stabilized triple network for *all* nodes is globally (not just pairwise) consistent. This requires the patchwork property: the set of all stabilized atomic relations forms a path-consistent atomic RCC5 network, which by [5] is globally consistent.

9 Conclusion

We have proposed a two-tier quotient construction for the finite representation of infinite $\mathcal{ALCI}_{RCC5}$ models:

1. **Algebraic foundations** (fully proven):
   - Reflexive PP is universally self-absorbing and fully composition-consistent with all external edges (Theorem 3.1, Corollary 3.2).
   - PP-transitivity forces strict linear order on distinct nodes, ruling out PP-cycles (Theorem 3.4).
   - External relations along PP-chains stabilize monotonically, with PP as the unique absorbing state (Lemma 4.6).
2. **Construction** (defined):
   - Period descriptors capture within-chain structure as finite cyclic words of Hintikka types (Definition 5.1).
   - PP-kernel nodes represent chains in the cross-chain RCC5 graph, with reflexive PP-loops.
   - Single-type chains collapse to one kernel node with full demand satisfaction (Theorem 5.6).
   - Multi-type chains require period descriptors (Theorem 5.8).
3. **Remaining gap** (Conjecture 7.1):
   - (P1) Every infinite PP-chain has an eventually periodic type sequence — plausible (finiteness of types + local constraints), proof sketch via Büchi automata.
   - (P2) Cross-chain kernel edges are composition-consistent — essentially follows from model consistency + stabilization, but needs careful formalization.

- (P3) Bounded quotient size — follows from (P1) + finiteness of type space.

If Conjecture 7.1 is true, combined with the soundness of the quasimodel approach [2] or an independent unfolding argument (Section 8), this yields decidability of $\mathcal{ALCI}_{RCC5}$. The two-tier quotient provides a concrete target for both the finite representation (completeness) and the model recovery (soundness), and its algebraic foundations are fully established.

The decidability of $\mathcal{ALCI}_{RCC5}$ remains open, but the two-tier quotient narrows the gap to a precise, well-defined conjecture about the periodic structure of PP-chains.

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